SOMAIYA VIDYAVIHAR UNIVERSITY

08/03/2025

Semester: Jan 2025 – Apr 2025 Maximum Marks: 30 Marks Examination: In-Semester Examination Duration : 1Hr:15 Min		
Programme code: 66 **Programme:** Honour Programme in Artificial Intelligence	**Class:** TY	**Semester:** VI (SVU 2020)
Name of the Constituent College: **K. J. Somaiya School of Engineering**	**Name of the department:** IT	
Course Code: 216h66C601	**Name of the Course:** Deep Learning	

Question No.		Max. Marks
Q1	Describe the steps of Backpropagation algorithm and its role in gradient based learning for training deep neural network.	10
Q2	What is activation function? Why is an activation function necessary? Explain any three activation functions in detail. Or Compare and Contrast different activation functions in terms of their advantages and disadvantages and impact on model performance.	10
Q3	Discuss the role of dropout, L1/L2 regularization, data augmentation and reducing overfitting. Or Explain the concept of Semi-Supervised learning. How does it differ from supervised and unsupervised learning?	10

1/1